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SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%
Assessment Tool Weightage: 25%

Duration: 90 minutes
Marks: 25

Code of Conduct:

I K. Lokesh Chandra (192425358) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Name of the student: K. Lokesh Chandra

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Total						

Signature of the Faculty:

Total Marks :

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques such as dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
 2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
 3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
 4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
 5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
 6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.
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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1: CNN FOR AUTOMATED MEDICAL IMAGE CLASSIFICATION

The healthcare organization wants to classify chest X-ray images into normal and disease-affected categories. CNN is suitable because it can automatically learn important visual features such as edges, textures and abnormal regions from images.

1. Suitable CNN Architecture

A suitable architecture is:

Input Image → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Convolution → ReLU → Fully Connected Layer → Output

The X-ray images can first be resized to a fixed size such as 224×224 pixels and normalized.

- Input layer: Receives the chest X-ray image.
- Convolution layers: Extract important features from the image.
- ReLU: Helps the network learn non-linear patterns.
- Pooling layers: Reduce the size of feature maps and computational cost.
- Fully connected layer: Combines the learned features.
- Output layer: Classifies the X-ray as normal or disease-affected.

The first convolution layers can learn simple features such as edges and lines. Deeper layers can learn more complex patterns related to abnormal regions.

2. Motivation and Role of CNN Components

CNN is suitable because medical X-ray images contain spatial patterns that can be automatically learned by the network.

Convolution Layer

The convolution layer uses filters to scan the image and extract useful features such as edges, shapes and textures.

Filters

Filters are small matrices that learn specific patterns. Different filters can detect different features.

Pooling Layer

Pooling reduces the size of feature maps while retaining important information. This reduces computation and helps the model focus on important features.

Fully Connected Layer

The fully connected layer combines the extracted features and uses them for final classification.

Output Layer

The output layer produces the final prediction:

- Normal
- Disease-affected

Thus, CNN can automatically learn features instead of requiring manual feature extraction.

3. Parameter Sharing

Parameter sharing means that the same filter is reused at different positions of an image. For example, a 3×3 filter has only 9 weights and one bias. The same filter can scan the complete X-ray image.

In a fully connected network, each pixel may have separate connections, resulting in a very large number of parameters.

Parameter sharing provides the following advantages:

- Reduces trainable parameters.
- Reduces memory usage.
- Reduces computational complexity.
- Makes CNN training faster.
- Allows the same feature to be detected at different positions.

Therefore, parameter sharing makes CNNs more efficient than fully connected networks for image classification.

4. Regularization Techniques

Overfitting occurs when the CNN performs well on training images but performs poorly on new X-ray images.

The following techniques can reduce overfitting:

Data Augmentation

Images can be slightly rotated, shifted, scaled or changed in brightness. This creates more varied training examples.

Dropout

Dropout randomly disables some neurons during training. This prevents the network from depending too much on particular neurons.

Batch Normalization

Batch normalization normalizes the values inside the network. It makes training more stable and can improve generalization.

Early Stopping

Training can be stopped when validation performance stops improving.

These techniques are especially useful because the case contains variations in image size, brightness and quality.

5. AlexNet or ResNet

Feature	AlexNet	ResNet
Depth	Relatively shallow	Much deeper
Feature learning	Good	Very good
Training	Easier	More complex

Feature	AlexNet	ResNet
Vanishing gradient	More possible	Reduced using skip connections
Accuracy	Good	Generally higher
Computation	Lower	Higher

Recommendation: ResNet

ResNet is more appropriate for this application.

ResNet can learn deeper and more complex features from X-ray images. Its residual/skip connections help reduce the vanishing-gradient problem and make deep networks easier to train. Although ResNet requires more computational resources than AlexNet, its stronger feature-learning capability and classification performance make it more suitable for medical image classification. The assessment specifically requires considering feature learning, network depth, computational requirements and classification performance.

2: CNN-BASED AUTONOMOUS VEHICLE OBJECT RECOGNITION

The autonomous vehicle must identify cars, pedestrians, traffic signs, bicycles and road obstacles under different lighting conditions, viewing angles, distances and road environments.

1. CNN Architecture Design

A suitable CNN architecture can be:

Camera Image → Preprocessing → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Deep Convolution Layers → Output

The camera captures road images and sends them to the CNN.

- The input layer receives the image.
- Convolution layers extract visual features.
- ReLU helps learn complex patterns.
- Pooling reduces feature-map size.
- Deeper layers learn complete object features.
- The output layer identifies the objects.

The model should be trained using images collected under different lighting, distances and road conditions so that it can work effectively in real-world situations.

2. Purpose of CNN Layers and Feature Learning

Input Layer

Receives images captured by the vehicle cameras.

Convolution Layer

Uses filters to detect features such as edges, shapes and textures.

Activation Layer

ReLU introduces non-linearity and allows the CNN to learn complex patterns.

Pooling Layer

Reduces the size of feature maps and lowers computational requirements.

Fully Connected/Output Layer

Uses the learned features to identify the object category.

Progressive Feature Learning

CNN learns features in stages:

Low-level features → Edges and lines

Middle-level features → Shapes and textures

High-level features → Cars, pedestrians, bicycles and signs

For example, the CNN may first identify edges, then shapes such as wheels and windows, and finally recognize the complete object as a car.

3. Importance of Parameter Sharing

CNN filters are shared across different locations of the image. The same filter can detect a particular feature whether it appears on the left, right, top or bottom of the image.

This is important for autonomous vehicles because objects can appear at different positions in camera images.

Advantages include:

- Fewer trainable parameters.
- Lower computational complexity.
- Lower memory requirements.
- Faster processing.
- Better detection of features at different positions.

Therefore, parameter sharing helps the vehicle recognize objects even when their locations change.

4. Overfitting and Regularization

Overfitting can occur when the model learns the training road images too closely and fails to recognize new environments.

This can be reduced using:

Data Augmentation

Create variations using changes in brightness, scaling, cropping, rotation and position.

Dropout

Randomly removes some neurons during training to reduce dependence on specific neurons.

Batch Normalization

Improves training stability and can improve generalization.

Early Stopping

Stops training when validation performance stops improving.

L2 Regularization

Controls very large weights and helps the model generalize to new images.

These techniques help the CNN perform reliably on unseen road conditions.

5. Comparison of AlexNet and ResNet

Factor	AlexNet	ResNet
Architecture	Traditional deep CNN	CNN with residual connections
Depth	Shallower	Much deeper
Training	Relatively easier	More complex
Vanishing gradient	More likely in deeper networks	Reduced by skip connections
Feature learning	Good	Excellent for complex features
Accuracy	Good	Generally better
Computational cost	Lower	Higher
Complex applications	Less suitable	More suitable

Analysis

AlexNet is simpler and requires fewer computational resources. However, ResNet can use a much deeper architecture and learn more complex features.

The residual connections in ResNet allow information to bypass some layers. This makes training deep networks easier and reduces the vanishing-gradient problem.

6. Recommended Architecture

Recommendation: ResNet

ResNet is the more suitable choice for the autonomous vehicle system.

The main reasons are:

1. It provides strong feature-learning capability.
2. It can identify complex objects and road features.
3. It handles deep networks more effectively.
4. Skip connections reduce the vanishing-gradient problem.
5. It can provide better classification performance.

However, autonomous vehicles require real-time processing, so computational cost is important. A suitable and optimized ResNet version should be selected to balance accuracy and processing speed.

Therefore, ResNet is recommended when high accuracy and robust object recognition are more important, while the model should be optimized to meet real-time requirements. This directly addresses the assessment requirement to justify the choice using accuracy, computational cost and real-time requirements.

CONCLUSION

CNNs are highly suitable for both medical image classification and autonomous vehicle vision because they automatically learn useful image features.

For the medical X-ray application, ResNet is preferred because it can learn complex features and provide strong classification performance.

For the autonomous vehicle application, ResNet is also recommended because it provides deeper feature learning and better handling of complex visual conditions. However, computational efficiency must be considered for real-time operation.

Overall, the proposed solutions demonstrate understanding of CNN architecture, convolution, filters, pooling, parameter sharing, regularization, AlexNet and ResNet, which are the main concepts covered by CO3.